Recency-Aware Multimodal Fusion for Irregular Time Series: Extending IMM-TSF from Forecasting to Classification

Brandi Jones, Chad Bishop, and Nnenna Ifeanyi

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1 Introduction

Multivariate time series data are ubiquitous across many domains such as healthcare, biology, and finance. These data rarely appear as clean, fully observed, regularly sampled sequences. Instead, they are often irregular, asynchronous, multimodal, and missing-rich, making them difficult to model using standard deep learning pipelines designed for regularly spaced tabular or sequential data. Additionally, the modality of irregularity, state, and quality of predictive information is important. Multimodal time-series learning approaches have been proposed in benchmark work such as Time-MMD, which emphasizes the need to incorporate information beyond numerical sequences alone [7].The need for models that can combine structured temporal signals with text while accounting for irregular sampling and modality mismatch has been the motivating factor for numerous prior studies [1, 5, 2, 3].

A central weakness of standard multimodal fusion is the fact that modern communication systems increasingly leverage multiple information streams, and the model may give high weight to a modality simply because its content appears predictive, even when it is "temporally misaligned, sparse, or unreliable [10]." In both clinical and surveillance domains, text and structured signals are not equally trustworthy for every sample. Raw textual data, in particular, often contains large portions of irrelevant information and potential data contamination, and require rigorous filtering processes to ensure data quality [7]. Temporal alignment between textual and numerical data is a requirement for most current time series frameworks, presenting yet another obstacle for real-world data analyses. Synchro-

nization of reported times with numerical time steps can be achieved with the use of binary timestamping, marking start and end dates across a universal temporal alignment, or with timestamp-to-text fusion (TTF) methods, such as Recency-Weighted Averaging (RecAvg) and Time2Vec-Augmented Cross-Attention (T2V-XAttn), which create time-aware text representations that help determine which past text should represent the current forecast query time [1]. These TTF methods enable enhanced integration of text contextual information with numerical time series data, but there is much room for improvement in terms of how strongly that text should influence the forecast once it is available [7].

Chang et al. (2025) developed a benchmark dataset and framework for time series modeling in the presence of irregularity and multimodality, Time-IMM/IMM-TSF. This framework allows for multivariate time-series forecasting that specifically addresses real-world irregularity mechanisms such as trigger-based, artifact-based, and constraint-based sampling. IMM-TSF also introduced multimodal fusion components designed to combine irregular numeric streams with aligned textual data. The purpose of this study was to extend the IMM-TSF framework in two ways: first, we propose a new reliability-aware gated residual additive multimodal fusion module (RA-GRAdd) that adjusts text-time-series fusion using recency, alignment, density, and missingness information. This extension allows the fusion step to depend not only on modality content, but also on whether the underlying modalities were temporally recent, temporally aligned, densely populated, and sufficiently observed. By modifying the gating function to include explicit reliability indicators, this project makes multimodal fusion more statistically and practically plausible for irregular real-world data. Second, our study extends the original forecasting-only IMM-TSF framework into binary classification settings, specifically modeling mortality prediction in MIMIC-IV critical care data and binary financial direction prediction in FNSPID market data.

2 Methods

2.1 Data Description

In an effort to reproduce the work of Chang et al.(2025) and strengthen the methodological credibility of the proposed extensions, we first benchmarked performance on multiple TIME-IMM datasets using the IMM-TSF framework, including GDELT, FNSPID, ClusterTrace, CESNET, and MIMIC-IV. Following reproduction efforts, our RA-GRAdd and classification extensions were applied to healthcare and financial domain time series datasets. The

healthcare application used MIMIC-IV, a large deidentified clinical database containing ICU and emergency department data with both structured measurements and free-text notes [6]. MIMIC-IV is a widely used public resource for clinical prediction research, with more than 299,000 patients and more than 431,000 ICU stays at Beth Israel Deaconess Medical Center (2008-2019). The financial application used FNSPID, a dataset combining 29.7 million stock price records, and 15.7 million time-aligned financial news records covering 4,775 S and P 500 companies [4].

2.2 Extension Problem Formulation

Let each sample i contain a structured irregular time-series branch and a text branch. In the proposed forecasting and classification extensions, each admission or entity is represented by three components:

$$\big(x_i,\, e_i,\, r_i\big), \tag{1}$$

where x_i is the structured feature representation, e_i is the note-derived or text-derived representation, and r_i is a four-dimensional reliability vector:

$$r_i = \begin{bmatrix} r_{i,\text{rec}} \\ r_{i,\text{align}} \\ r_{i,\text{dens}} \\ r_{i,\text{miss}} \end{bmatrix}. \tag{2}$$

The first extension preserves the original forecasting setting. Let the structured time-series branch for sample i be

$$X_i^{(\text{ts})} = \{(t_{ij}, x_{ij}) : j = 1, \ldots, m_i\}, \tag{3}$$

where t_{ij} is the irregular observation time and $x_{ij} \in \mathbb{R}^q$ is the multivariate observation vector at that time. Let the associated text branch be

$$D_i = \{(s_{ik}, d_{ik}) : k = 1, \ldots, n_i^{(\text{txt})}\}, \tag{4}$$

where s_{ik} is the text timestamp and d_{ik} is the text content or a precomputed embedding.

After preprocessing and windowing, each forecasting sample is converted into

$$X_i^{(\text{hist})} \in \mathbb{R}^{L_i \times q}, \quad M_i^{(\text{hist})} \in \{0,1\}^{L_i \times q}, \quad t_i^{(\text{hist})} \in \mathbb{R}^{L_i}, \tag{5}$$

together with a prediction target over the forecast horizon and a text representation

$$E_i^{(\mathrm{txt})} \in \mathbb{R}^{K_i \times p_{\mathrm{txt}}}. \tag{6}$$

Given a history window of length H and a prediction window of length P, the forecasting objective is to learn a function f_Θ such that

$$\hat{Y}_i = f_\Theta\Big(X_i^{(\mathrm{hist})}, M_i^{(\mathrm{hist})}, t_i^{(\mathrm{hist})}, E_i^{(\mathrm{txt})}\Big), \tag{7}$$

with mean squared forecasting loss

$$L_{\mathrm{MSE}} = \frac{1}{N}\sum_{i=1}^{N} \|Y_i - \hat{Y}_i\|_2^2. \tag{8}$$

The second extension adapts the framework to binary classification. Let $y_i \in \{0, 1\}$ denote the binary class label, such as mortality or market direction. The classification objective is to learn a function

$$\hat{p}_i = g_\Theta(x_i, e_i, r_i), \tag{9}$$

where $\hat{p}_i = P(y_i = 1 \mid x_i, e_i, r_i)$ is the predicted class probability. In this way the multimodal fusion framework can be extended from forecasting to classification.

2.3 Structured and Text Representations

The structured branch is encoded into a latent vector

$$h_i^{(\mathrm{ts})} = \phi_{\mathrm{ts}}\Big(X_i^{(\mathrm{hist})}, M_i^{(\mathrm{hist})}, t_i^{(\mathrm{hist})}\Big) \in \mathbb{R}^d. \tag{10}$$

In the forecasting experiments, the primary structured backbone followed the tPatchGNN-based setup used in the motivating framework [1, 11]. In the classification extension, the structured history was summarized into a fixed-dimensional representation suitable for patient-level or entity-level prediction.

The text branch is encoded into a latent vector

$$h_i^{(\mathrm{txt})} = \phi_{\mathrm{txt}}\Big(E_i^{(\mathrm{txt})}\Big) \in \mathbb{R}^d. \tag{11}$$

In practice, text was represented using precomputed embeddings, including GPT-2-based note features in the implementation pipeline. Both branches are projected into a common latent fusion space:

$$\tilde{h}_i^{(\mathrm{ts})} = \rho\Big(W_{\mathrm{ts}} h_i^{(\mathrm{ts})} + b_{\mathrm{ts}}\Big), \tag{12}$$

$$\tilde{h}_i^{(\mathrm{txt})} = \rho\Big(W_{\mathrm{txt}} h_i^{(\mathrm{txt})} + b_{\mathrm{txt}}\Big), \tag{13}$$

where $\rho(\cdot)$ denotes a nonlinear activation, such as ReLU, optionally followed by dropout.

2.4 Original Gated Fusion

In the original IMM-TSF-style multimodality fusion module, the structured and text representations are concatenated [1]:

$$c_i = \begin{bmatrix} \tilde{h}_i^{(\mathrm{ts})} \\ \tilde{h}_i^{(\mathrm{txt})} \end{bmatrix}. \tag{14}$$

A content-only gate is then computed as

$$a_i = W_{g2}\, \rho(W_{g1} c_i + b_{g1}) + b_{g2}, \tag{15}$$

followed by

$$g_i^{(\mathrm{old})} = \sigma(a_i), \tag{16}$$

where $\sigma(\cdot)$ is the elementwise sigmoid function. The fused representation is

$$z_i^{(\mathrm{old})} = g_i^{(\mathrm{old})} \odot \tilde{h}_i^{(\mathrm{ts})} + \Big(1 - g_i^{(\mathrm{old})}\Big) \odot \tilde{h}_i^{(\mathrm{txt})}. \tag{17}$$

This gate learns how to weight structured and text signals based only on feature content.

2.5 Proposed RA-GRAdd Extension

The first key contribution of this project is the reliability-aware residual additive gate (RA-GRAdd). Rather than relying solely on content, the model computes a reliability vector

$$r_i = \begin{bmatrix} r_{i,\text{rec}} \\ r_{i,\text{align}} \\ r_{i,\text{dens}} \\ r_{i,\text{miss}} \end{bmatrix}, \tag{18}$$

with the four components corresponding to recency, text-to-time series alignment, note density, and missingness.

2.5.1 Recency

Let $t_i^{(\text{query})}$ denote the forecast or decision time, and let $\bar{s}_i$ denote a representative time for the available text observations. Then recency reliability is modeled as an exponential decay in time distance:

$$r_{i,\text{rec}} = \exp\left(-\frac{|t_i^{(\text{query})} - \bar{s}_i|}{\tau_{\text{rec}}}\right), \tag{19}$$

where $\tau_{\text{rec}} > 0$ is a decay hyperparameter.

2.5.2 Alignment

The alignment term is defined by comparing the midpoint of the structured observation timeline to the midpoint of the text timeline. Let

$$\bar{t}_i^{(\text{ts})} = \frac{t_{i,\min}^{(\text{ts})} + t_{i,\max}^{(\text{ts})}}{2}, \qquad \bar{t}_i^{(\text{txt})} = \frac{s_{i,\min}^{(\text{txt})} + s_{i,\max}^{(\text{txt})}}{2}, \tag{20}$$

and define temporal misalignment as

$$\delta_i = \left|\bar{t}_i^{(\text{ts})} - \bar{t}_i^{(\text{txt})}\right|. \tag{21}$$

Then alignment reliability is

$$r_{i,\text{align}} = \exp\left(-\frac{\delta_i}{\tau_{\text{align}}}\right). \tag{22}$$

In our implementation, $\tau_{\text{align}} = 72$ hours was used as a practical value.

2.5.3 Density

Let the time-series note/text density be

$$\text{density}_i = \frac{m_i}{\max(\text{timespan}_i, 1)}. \tag{23}$$

A bounded density reliability score is then

$$r_{i,\text{dens}} = \min(\text{density}_i, 1). \tag{24}$$

2.5.4 Missingness

Let M_{ijk} be a missingness indicator for variable k at time point j:

$$M_{ijk} = \begin{cases} 1, & \text{if variable } k \text{ is missing at time } j, \\ 0, & \text{otherwise.} \end{cases} \tag{25}$$

The average missingness rate is

$$\bar{M}_i = \frac{1}{m_i q} \sum_{j=1}^{m_i} \sum_{k=1}^{q} M_{ijk}, \tag{26}$$

and the missingness reliability is

$$r_{i,\text{miss}} = 1 - \bar{M}_i. \tag{27}$$

2.5.5 Reliability Projection

The reliability vector is mapped into the same latent fusion dimension:

$$u_i = \psi(r_i) = W_{r2}\, \rho(W_{r1} r_i + b_{r1}) + b_{r2}, \tag{28}$$

where $u_i \in \mathbb{R}^d$ is a learned reliability correction.

2.5.6 Reliability-Aware Residual Additive Gate

The content-based pre-sigmoid score remains

$$a_i = W_{g2}\,\rho(W_{g1}c_i + b_{g1}) + b_{g2}. \tag{29}$$

The proposed gate adds the reliability correction before the sigmoid:

$$g_i^{(\mathrm{RA})} = \sigma(a_i + u_i). \tag{30}$$

Equivalently, for latent dimension ℓ,

$$g_{i\ell}^{(\mathrm{RA})} = \sigma(a_{i\ell} + u_{i\ell}). \tag{31}$$

The fused representation becomes

$$z_i^{(\mathrm{RA})} = g_i^{(\mathrm{RA})} \odot \tilde{h}_i^{(\mathrm{ts})} + \left(1 - g_i^{(\mathrm{RA})}\right) \odot \tilde{h}_i^{(\mathrm{txt})}. \tag{32}$$

This innovation changes the gating question from "what should be weighted based on content alone?" to "what should be weighted based on content, corrected by how trustworthy that content is?" The original IMM-TSF model is nested within the proposed extension formulation. If the reliability correction is identically zero, then

$$g_i^{(\mathrm{RA})} = \sigma(a_i + 0) = \sigma(a_i) = g_i^{(\mathrm{old})}. \tag{33}$$

2.6 Classification Head and Loss

In classification pipelines, after fusion the latent vector is fed into a multilayer perceptron classifier:

$$\eta_i = W_o z_i + b_o, \qquad \hat{p}_i = \sigma(\eta_i). \tag{34}$$

The model is trained using weighted binary cross-entropy loss to address class imbalance:

$$\mathcal{L}_{\mathrm{WBCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_+ y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\right], \tag{35}$$

where w_+ is the positive-class weight estimated from the training set. In the classification setting, the experiments used standard 60/20/20 train/validation/test splits and evaluated

discrimination and classification performance using metrics such as AUROC, AUPRC, accuracy, precision, recall, and F1 score.

2.7 Ablation Design

In keeping with the benchmark framework formulated by Chang et al. (2025), an ablation framework was used for assessment of the RA-GRAdd gating extension. The ablation framework separates the full reliability gating idea into four individual reliability mechanisms to determine whether performance gains come from the overall concept or from one specific ingredient.

Let the full reliability vector be

$$r_i = [r_{i,\text{rec}},\, r_{i,\text{align}},\, r_{i,\text{dens}},\, r_{i,\text{miss}}]^\top. \tag{36}$$

For ablation mask $m^{(a)} \in \{0,1\}^4$, define the masked reliability vector

$$\tilde{r}_i^{(a)} = m^{(a)} \odot r_i, \tag{37}$$

and the corresponding correction

$$u_i^{(a)} = \psi\left(\tilde{r}_i^{(a)}\right). \tag{38}$$

Then the ablated gate is

$$g_i^{(a)} = \sigma(a_i + u_i^{(a)}). \tag{39}$$

The implemented variants were:

$$\text{GR Add:} \quad g_i^{(\text{old})} = \sigma(a_i), \tag{40}$$

$$\text{RA full:} \quad g_i = \sigma\big(a_i + \psi([r_{\text{rec}}, r_{\text{align}}, r_{\text{dens}}, r_{\text{miss}}]^\top)\big)\,, \tag{41}$$

$$\text{RA recency:} \quad g_i = \sigma\big(a_i + \psi([r_{\text{rec}}, 0, 0, 0]^\top)\big)\,, \tag{42}$$

$$\text{RA alignment:} \quad g_i = \sigma\big(a_i + \psi([0, r_{\text{align}}, 0, 0]^\top)\big)\,, \tag{43}$$

$$\text{RA density:} \quad g_i = \sigma\big(a_i + \psi([0, 0, r_{\text{dens}}, 0]^\top)\big)\,, \tag{44}$$

$$\text{RA missingness:} \quad g_i = \sigma\big(a_i + \psi([0, 0, 0, r_{\text{miss}}]^\top)\big)\,. \tag{45}$$

2.8 Training Procedure

Algorithmically, the forecasting workflow was:

1. Read processed record-level folders from the dataset-specific processed directory.
2. Build classification or forecasting windows from the specified history and prediction spans.
3. Read the precomputed text embedding file in each record directory.
4. Construct structured tensors and note embeddings.
5. Compute one of the six gate variants.
6. Train the same classification or forecasting model with the same split structure.
7. Save comparison tables and plots to disk.

This design ensured that performance differences could be attributed to the fusion strategy rather than to changes in datasets, output heads, or split definitions.

3 Results

3.1 Benchmark Reproduction Results

Table 1 summarizes our original IMM-TSF model reproduction comparisons. Across datasets, reproduced forecasting errors were generally close to those reported in the motivating paper, although discrepancies varied by dataset and irregularity type.

Table 1: Model performance comparisons for reproduction results from Chang et al. across TIME-IMM benchmark datasets. Lower MSE indicates better performance.

Dataset	Irregularity Type	Multimodal MSE		Unimodal MSE	
		Paper	Ours	Paper	Ours
GDELT	Trigger-Based (Event-Based Logging)	1.03	1.02	1.03	1.04
FNSPID	Artifact-Based (Operational Window Sampling)	0.12	0.16	0.12	0.16
ClusterTrace	Artifact-Based (Resource-Aware Collection)	0.61	0.85	0.98	0.87
CESNET	Constraint-Based (Scheduling Jitter / Delay)	0.91	0.96	0.98	1.03
MIMIC-IV	Trigger-Based (Event-Based Logging)	0.77	0.74	0.73	0.74

3.2 Forecasting Extension Results

3.2.1 MIMIC Forecasting

The MIMIC forecasting experiments used 562 total windows, including 359 training windows, 90 validation windows, and 113 test windows. The structured input dimension was 14 and the text embedding dimension was 768. RA alignment achieved the best overall performance on MIMIC, reducing test MSE from 1118.1240 under GR-Add to 1064.9611, an improvement of approximately 4.75%, as shown in Table 2.

Table 2: Forecasting comparison for MIMIC. Lower values indicate better performance.

Gate	Total	Train	Val	Test	Input Dim	Emb Dim	Val MSE	Test MSE
GR Add	562	359	90	113	14	768	3483.120809	1118.124023
RA full	562	359	90	113	14	768	3457.153137	1100.617920
RA recency	562	359	90	113	14	768	3476.335312	1116.817383
RA alignment	562	359	90	113	14	768	3435.766754	1064.961060
RA density	562	359	90	113	14	768	3490.095149	1136.723755
RA missingness	562	359	90	113	14	768	3474.737991	1109.925537

Gate	Test MAE
GR Add	25.277424
RA full	25.436848
RA recency	25.688364
RA alignment	24.709850
RA density	26.040966
RA missingness	25.438681

3.2.2 FNSPID Forecasting

The FNSPID forecasting experiments used 80 total windows, including 51 training windows, 13 validation windows, and 16 test windows. The structured input dimension was 6 and the text embedding dimension was 32. Because the initial scale of the target values produced unstable raw loss magnitudes, normalized reporting was also examined. By test MSE, RA full performed best on FNSPID, improving over GR-Add from 1.165095×10^{13} to 1.066716×10^{13}, corresponding to an improvement of approximately 8.44% (Table 3, 4). However, GR Add retained a slightly smaller MAE, as shown in Table 3.

Table 3: Normalized forecasting comparison for FNSPID. Lower values indicate better performance.

Gate	Total	Train	Val	Test	Input Dim	Emb Dim	Val MSE	Test MSE
GR Add	80	51	13	16	6	32	1.050019	1.165095×10^{13}
RA full	80	51	13	16	6	32	1.086179	1.066716×10^{13}
RA recency	80	51	13	16	6	32	1.200649	1.393435×10^{13}
RA alignment	80	51	13	16	6	32	1.486357	1.241561×10^{13}
RA density	80	51	13	16	6	32	1.559003	1.275653×10^{13}
RA missingness	80	51	13	16	6	32	1.064489	1.405441×10^{13}

Gate	Test MAE
GR Add	1099964.750
RA full	1105815.500
RA recency	1297888.750
RA alignment	1238156.000
RA density	1317977.125
RA missingness	1286875.875

3.2.3 Cross-Dataset Summary

Table 4: Best gate by dataset. Lower values indicate better performance.

Dataset	Best Gate	Test MSE	Test RMSE	Test MAE	Improve on GR-Add
MIMIC	RA alignment	1064.961060	32.633741	24.709850	4.75%
FNSPID	RA full	1.066716×10^{13}	3.266062×10^{6}	1.105816×10^{6}	8.44%

3.3 Classification Results

IMM-TSF was successfully extended from irregular multimodal forecasting to binary classification in both healthcare and financial settings while preserving interpretability and improving fusion through reliability-aware gating. In the MIMIC-IV mortality application, the final analytic cohort included $N = 1108$ adult single-admission ICU stays with length of stay greater than 30 days and sufficient structured and text data to construct the multimodal

representation. In the financial application, the processed FNSPID classification subset included $N = 678$ usable multimodal samples. Across both domains, the original forecasting decoder was replaced with a binary classification head trained using binary cross-entropy loss with held-out train, validation, and test splits. In the mortality task, the strongest model was the RA-GRAdd variant excluding missingness, which achieved an AUC of 0.90, AUPRC of 0.82, balanced accuracy of 0.89, precision of 0.83, recall of 0.63, and F1 score of 0.71. In the financial task, the strongest overall discrimination was observed for RA-GRAdd with all four reliability components, which achieved an AUC of 0.81 and AUPRC of 0.80, while the missingness-ablated variant yielded the highest accuracy and F1 score at 0.80 and 0.76, respectively (Table 5, 6). The ROC curves and confusion matrices further showed that reliability-aware variants provided stronger class separation than the baseline gate alone, and permutation importance analyses indicated that both structured variables and text-derived embeddings contributed meaningfully to prediction, supporting the value of reliability-aware text integration rather than naive late fusion (Figure 1-5).

Table 5: Final mortality classification model results on the MIMIC-IV cohort. Higher values indicate better performance.

Model	AUC	AUPRC	Bal. Accuracy	Precision	Recall	F1	Threshold
RA-GRAdd (-Missingness)	0.90	0.82	0.89	0.83	0.63	0.71	0.50
RA-GRAdd (-Recency)	0.88	0.81	0.89	0.81	0.66	0.72	0.51
RA-GRAdd (-Density)	0.90	0.81	0.87	0.79	0.67	0.69	0.57
RA-GRAdd (+All 4)	0.89	0.79	0.87	0.77	0.67	0.68	0.65
Original GR-Add	0.88	0.79	0.87	0.77	0.63	0.70	0.49
RA-GRAdd (-Alignment)	0.87	0.77	0.84	0.70	0.55	0.62	0.61

Table 6: Final financial classification model results on the processed FNSPID subset. Higher values indicate better performance.

Model	AUC	AUPRC	Accuracy	F1	Precision	Recall	Threshold
RA-GRAdd (+All 4)	0.81	0.80	0.77	0.76	0.77	0.75	0.31
RA-GRAdd (-Missingness)	0.80	0.77	0.80	0.76	0.76	0.77	0.31
RA-GRAdd (-Recency)	0.80	0.77	0.75	0.77	0.77	0.77	0.28
RA-GRAdd (-Alignment)	0.80	0.77	0.77	0.75	0.77	0.79	0.32
Original GR-Add	0.79	0.75	0.74	0.74	0.70	0.78	0.13
RA-GRAdd (-Density)	0.78	0.75	0.75	0.71	0.78	0.66	0.30

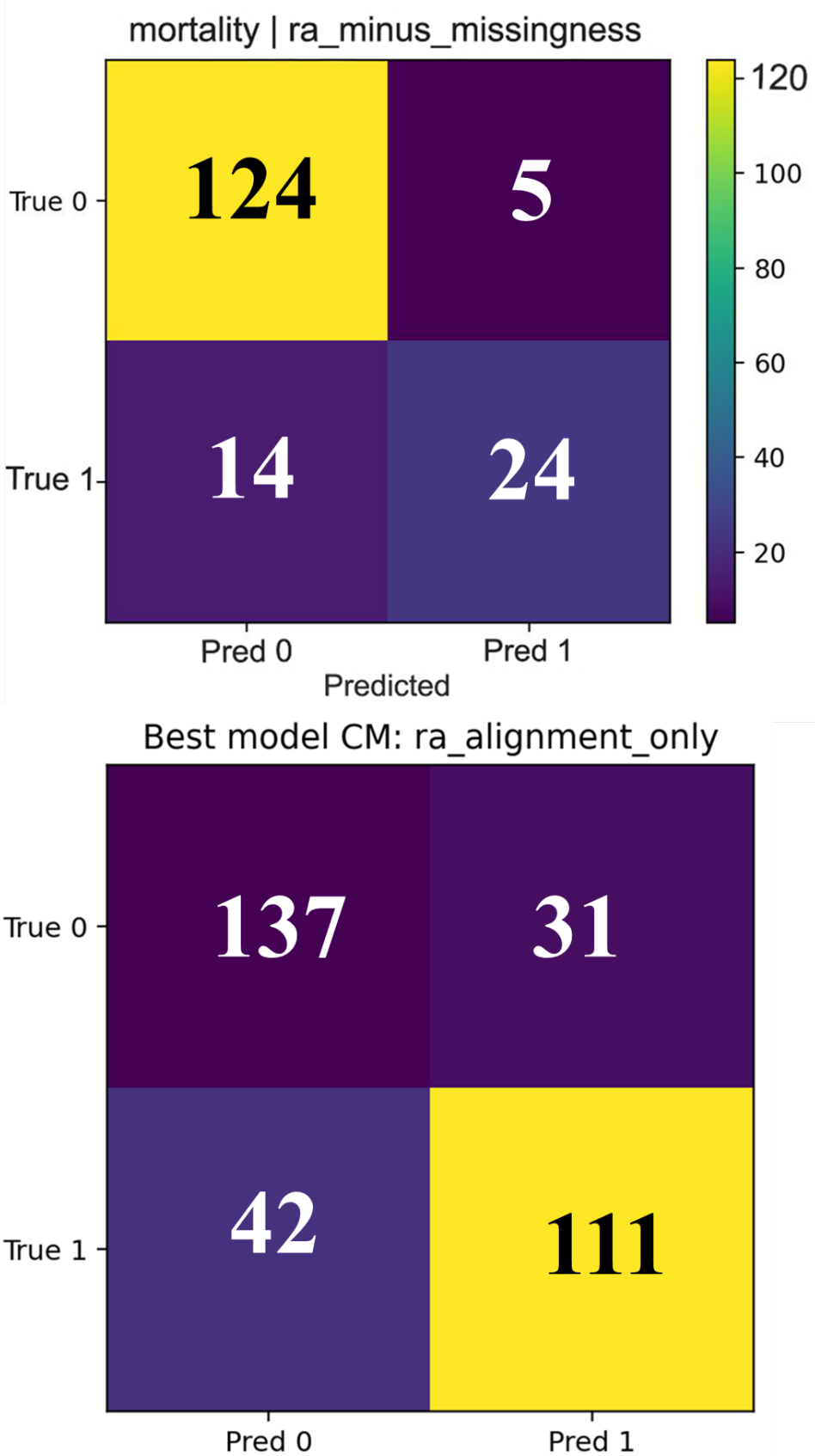

Figure 1: Confusion matrices for the final mortality and financial market direction classification models. The reliability-aware variants show stronger separation of positive and negative classes than the baseline gated residual addition models.

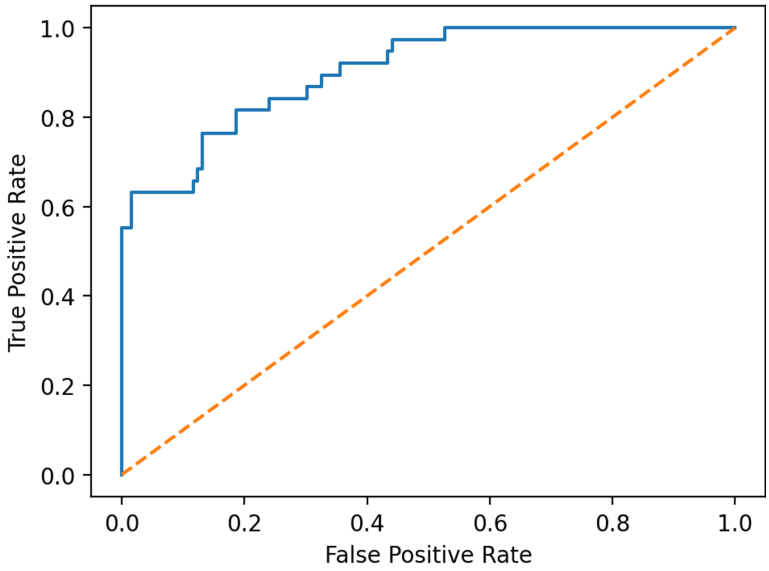

Figure 2: Receiver operating characteristic curve for the best-performing mortality classification model, RA-GRAdd without missingness. The test AUROC was 0.90.

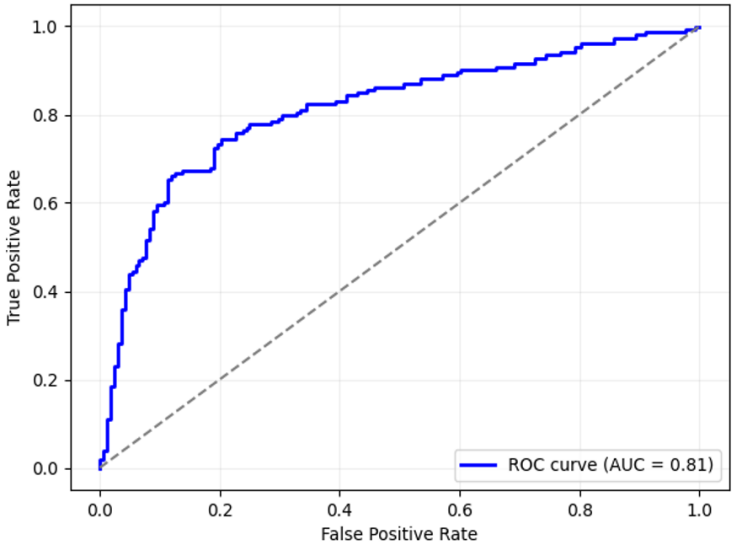

Figure 3: Receiver operating characteristic curve for the best-performing financial classification model, RA-GRAdd with all four reliability components. The test AUROC was 0.81.

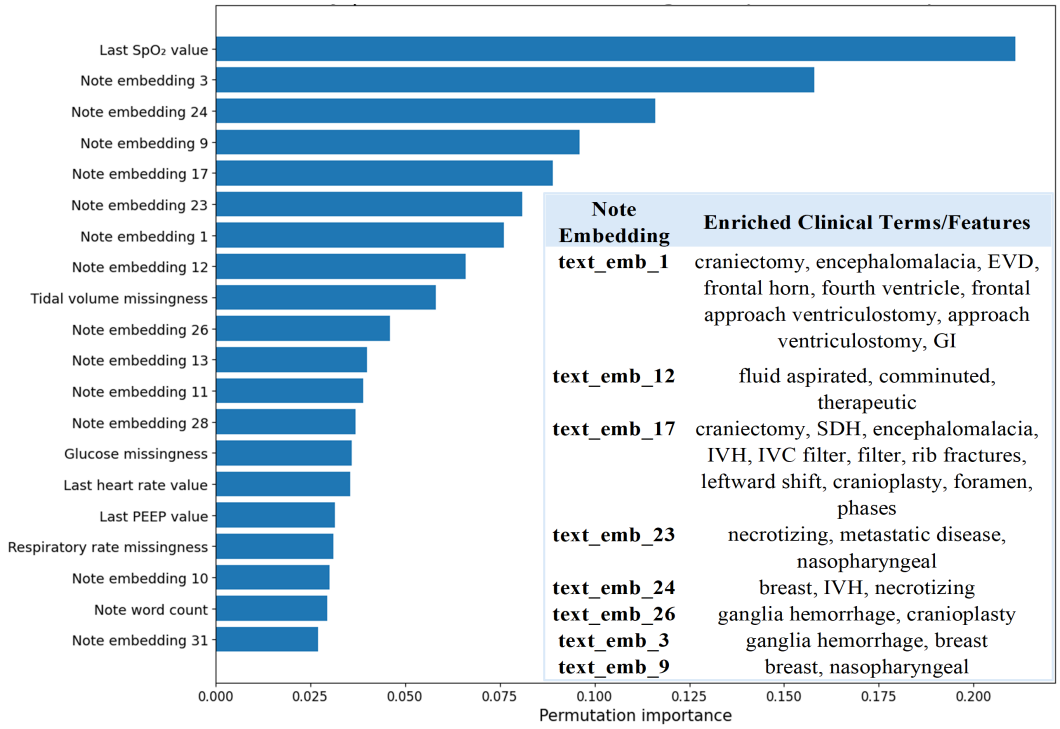

Figure 4: Permutation importance for the final mortality classification model. Both structured features and note-embedding features contributed meaningfully to prediction, supporting the value of multimodal reliability-aware fusion.

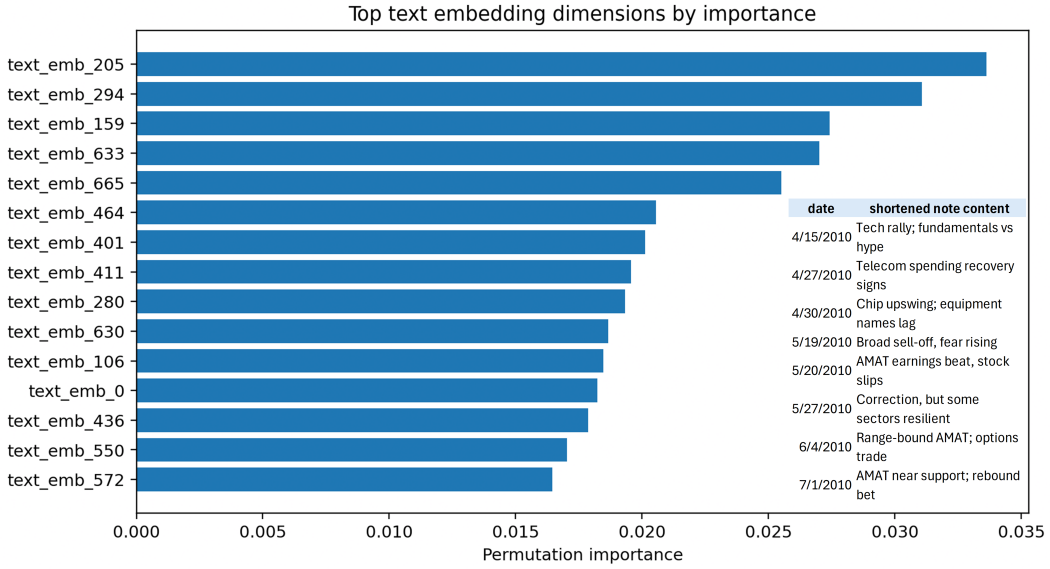

Figure 5: Permutation importance for the final financial classification model. The importance profile suggests that reliability-aware fusion preserved useful contributions from both structured market signals and text-derived features.

4 Discussion

Our findings support the conclusion that multimodal fusion should depend not only on feature content, but also on whether the contributing modalities are trustworthy at the time of prediction. The proposed RA-GRAdd mechanism makes this idea explicit by incorporating recency, note density, text–time-series alignment, and missingness into the fusion process before combining structured and text information. Rather than treating all available text as equally informative, the model calibrates how much trust should be assigned to asynchronous textual information relative to the structured time-series branch. In this sense, reliability-aware fusion enables the model to account for modality quality, timing, and completeness before performing the final integration step.

The forecasting and classification results suggest that the most informative reliability signal may be dataset-specific. In MIMIC forecasting, the strongest improvement was obtained with the alignment-only gate, indicating that the temporal correspondence between clinical notes and structured observations was the most useful reliability cue in that setting. In contrast, for FNSPID forecasting, the full reliability-aware gate achieved the best performance by test MSE, suggesting that multiple reliability dimensions jointly contributed to improved prediction in the financial setting. Taken together, these findings indicate that the proposed framework is flexible enough to adapt to different real-world irregularity mechanisms rather than overcommitting to a single fusion pattern.

Additionally, IMM-TSF framework was shown to extend successfully beyond forecasting to downstream binary classification. The classification experiments demonstrated that irregular multimodal forecasting architectures can be adapted to classification tasks by replacing the forecasting decoder with a binary prediction head while preserving the same core reliability-aware fusion concept. Across both healthcare and financial applications, reliability-aware gating improved the usefulness of asynchronous text while maintaining interpretability through ROC curves, confusion matrices, and permutation importance analyses (Figures 1-5). More broadly, these results suggest that improved handling of irregularity and cross-modal information quality can strengthen not only forecasting performance, but also classification performance, in settings where structured and textual data arrive at different times and with different levels of completeness.

5 Limitations

This study has several limitations. First, some datasets, particularly FNSPID, yielded relatively small effective sample sizes after preprocessing and window construction, which may limit the stability and generalizability of some results. Second, the proposed classification extension depends in part on the quality of temporal alignment between text and structured time-series data, and this alignment may vary substantially across domains and preprocessing pipelines. Third, although the reliability-aware components were motivated by real-world multimodal data characteristics, the present study evaluated a limited set of reliability measures, and additional formulations may be useful in future work. Finally, further evaluation is needed across broader task settings, larger datasets, and additional outcome definitions to determine how consistently these reliability-aware mechanisms improve performance in both forecasting and classification tasks.

6 Conclusion

This study introduces a methodological extension to IMM-TSF [1] through a reliability-aware residual additive gate that transforms the original content-only gate into a reliability-aware fusion mechanism:

$$g_i^{(\mathrm{old})} = \sigma(a_i), \qquad g_i^{(\mathrm{RA})} = \sigma(a_i + u_i), \tag{46}$$

where u_i is a learned projection of recency, alignment, density, and missingness information. This extension preserves the core spirit of IMM-TSF, while improving both the scientific plausibility and empirical performance of multimodal fusion for irregular, asynchronous, and missing-rich time-series learning. Across the forecasting experiments, the proposed extension improved on the original GR-Add baseline in both datasets. The best-performing MIMIC forecasting model was RA alignment, whereas the best-performing FNSPID forecasting model by test MSE was RA full. More broadly, this project demonstrates that IMM-TSF can be extended from forecasting to classification while retaining interpretability, modularity, and compatibility with real-world multimodal time-series data. As with forecasting, the RA-GRAdd extension improved classification with the RA minus missingness model performing best overall in MIMIC mortality classificaiton, and the RA-GRAdd full model performing best in FNSPID financial market direction prediction.

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